

IX AND TLEILAX

When it comes to technology, most Houses within the Imperium obey the edicts of the Great Convention and adhere to the edicts of both the Butlerian Jihad and the Emperor's own will. Two worlds and their inhabitants, however, buck this trend and skirt the edge of violating these proscriptions.

IX

Ix, the ninth planet in its star system, is named for its number (IX), and is home to House Vernius, a Major House. The inhabitants, Ixians, are famed particularly for technological innovation. Ixian devices are prized throughout the Imperium, despite the degree to which the Ixians are thought to disregard the Butlerian Edicts about the development of thinking machines. Many servos and other such cunning creations come from Ix, and it is suspected that their advances in technology go far beyond what they reveal and are willing to trade.

In the recent past, the ruling body, sometimes called the Ixian Confederacy, engaged in fierce competition with House Richese, another rival in the field of machinery and technological development. Richese suffered mightily in this conflict and has yet to recover fully.

TLEILAX

No group of people is considered with more suspicion within the Imperium than the Bene Tleilax, the inhabitants of the planet Tleilax. An isolationist people who exist outside the Fauflreluches caste system, they are neither part of the Landsraad nor fall within the category of the Houses that populate it. In appearance, Tleilaxu look mostly human, with curious features and sharpened teeth.

However, the Tleilaxu are met with widespread revulsion throughout the Imperium, for their primary export is one that violates the spirit of the Butlerian Edicts, if not its precise wording. They can clone dead flesh, resurrecting the dead into gholas, living beings genetically identical to their original forms, if lacking in memory. They also are masters of artificial limbs and organs, a specialty being artificial eyes, which always look as if made of polished steel.

Another reason the Tleilaxu are so reviled and distrusted are their 'Face Dancers', genetic eunuch shape-changers able to manipulate their features and physiques to assume the form of others, making them natural spies, assassins, and operatives. Face Dancers operate on behalf of the Tleilaxu or, at times, as agents for hire. In disguise, they can infiltrate almost any House and cause untold damage. This adds to their untrustworthy reputation, as to outsiders, any Tleilaxu may be a Face Dancer, or, by extension, anyone at all. Only the Bene Gesserit have proven capable of detecting Face Dancers.

CHOAM

(COMBINE HONETTE OBER ADVANCER MERCANTILES)

While the Imperial House, the great Houses of the Landsraad, and the Spacing Guild seem the most obvious powers in the Imperium, all of them are beholden to and integrated with Combine Honette Ober Advancer Mercantiles (CHOAM). This corporation has existed for millennia and is the ultimate arbiter of economic power in the known universe. Every standard day, trillions upon trillions of solaris are transferred through CHOAM, and billions of lives are impacted. People live and die for the merest fraction of this incalculable wealth, and whole worlds have been scoured in the pursuit of the coveted directorship positions on CHOAM'S board. Through CHOAM, power is delivered to the rulers of the Imperium and those rulers are capable of anything to maintain their grasp upon it.

Established in the proceedings following the Great Convention, CHOAM's official function is the regulation and management of all interstellar trade. Thus, all commerce conducted via the Spacing Guild falls under CHOAM's purview. The sales of countless

goods and services are transacted under CHOAM's auspices; metals, lumber, pundi rice, whale fur, artworks, contracts securing the services of the Swordmasters of Ginaz, the machines of Ix and Richese, and of course, the spice melange of Arrakis, to name but a fraction of the vast array of goods, services, and personnel administered by CHOAM.

SHARES AND SHAREHOLDERS

CHOAM is a separate entity from the other institutions of the Imperium but is comprised of members of those powers. The Imperial House Corrino controls the largest block of shares in CHOAM and is empowered to assign and revoke directorships. These are invariably granted to the leaders of the Major Houses of the Landsraad, guided by the attitude the Emperor has towards them.

The remaining shares are owned by the thousands of Major and Minor Houses of the Landsraad, though



a substantial portion is held by the Spacing Guild itself, and the Bene Gesserit, who operate as silent partners. A small number of shares are controlled by private individuals and lesser organizations, typically awarded by the largesse of a House or the Emperor himself.

Shareholders are entitled to a commensurate share of CHOAM profits, from which all meaningful wealth in the universe is derived. While small private fortunes can be amassed from the operations of commerce on a single planet, these amounts pale in comparison to those generated by interstellar trade of any commodity. A land baroness on Chusuk may live a comfortable life but unless she also has her hand in the off-planet trade of her world's superlative musical instruments, her wealth and power are trivial by the standards of the greater universe. Accordingly, the wages earned by the common people of the Faufreluches are utterly inconsequential in this economic structure.

CHOAM shares and directorships are hereditary. Unless the Emperor himself chooses to intervene, the offspring of a shareholder or director can expect to inherit those responsibilities and benefits upon the death of their parents. Naturally, this is a leading motivation of intra-House intrigue as young nobles desire not just the inheritance of title, but wealth, or wish to see themselves land higher in the line of succession than their peers. Over the ages, many noble youths have fallen victim to poisoned food or drink—called *chaumurky* or *chaumas*—simply so their sibling does not need to split the shares to someday be inherited.

SHAREHOLDER RESPONSIBILITIES

Shareholder Houses typically control several valuable commodities in a portfolio or share control of several especially lucrative trades with partner Houses. A House Minor might oversee the export of exotic lumbers grown only on their homeworld, the contracting of a specialized labor force for use in off-world industries, and the manufacture of a prized class of racing ornithopter all as part of their portfolio of commodities. The generation of profit from the portfolio to the aggregated CHOAM holdings is the shareholder's primary responsibility. All duties necessary to maintain the production, export, and import of items relevant to their portfolio's trades are the responsibility of the shareholder. Houses that share control of a trade with partners must find ways to equitably split the cost of business: a common cause of strife between Houses.

Should a shareholder find a way to increase profits through efficient manufacturing or agricultural

processes they may find their general standing in CHOAM rise. Truly impressive innovations can potentially bring a shareholder into the spotlight and into consideration by the Emperor for a seat on the board of directors.

Shareholders are expected to provide regular financial reports and keep detailed accounting. Official meetings of CHOAM shareholders are infrequent and never mandatory. The vast size of the universe makes such meetings difficult to arrange at best, and thus most official CHOAM business is conducted via couriers transported by the Spacing Guild. Business meetings between shareholders with common interests are more common but these are conducted as part of standard day-to-day business.

Shareholders who are found in violation of CHOAM's operational rules will find themselves formally censured. Those who have been found committing minor acts of embezzlement or simple bad accounting will typically be fined or given a simple warning. More severe penalties may include garnished profits and oversight by outside parties appointed by the directorate. Those found to have embezzled significant funds, interfered negatively in the business of others, or otherwise committed grievous financial wrongdoing may find themselves facing more severe penalties, up to and including the loss of all shares and membership in CHOAM itself.

Provided their annual reports maintain a generally profitable status quo most shareholders can expect no interference in how they conduct their business. Directives from the board are uncommon but must be carried out by shareholders with haste and efficiency. However, should loss be incurred as a pattern, the shareholding House can expect to be questioned, investigated, and even audited. CHOAM auditors are as ruthless as those in service to the Emperor's tax collection and have left numerous Houses in financial ruin and at the mercy of their peers.

Particularly important and profitable commodities are entrusted only to those most loyal to the Imperial House. These most valuable interests represent a double-edged sword for those responsible, usually a House Major holding a directorship. Truly astonishing wealth is in the grasp of a House managing such a commodity but endangering a vital trade will incur the wrath of the Emperor, the other Houses of the Landsraad, and perhaps less obvious parties. Naturally, such fortunes attract the attentions of rivals who will do anything in their power to disrupt operations to see them fail. Sabotage, industrial espionage, and even assassination are prohibited, but not uncommon elements of doing business in CHOAM.

THE BOARD OF DIRECTORS

CHOAM'S directorate is one of the most powerful organizations in the Imperium. Seats on the board of directors are highly coveted as they carry with them not only the opportunity for stupendous wealth through control of prized commodities, but also the ear of the Emperor. Directors set economic policy, oversee vast trade networks, and are privy to financial information long before other shareholders.

The public face of CHOAM is the President, serving in the administrative building, the Silver Needle on Kaitain. The President, however, takes his final orders from the far more powerful CHOAM Ur-Director, who is rarely seen in public.

Directorships are granted by the Emperor, typically as a reward for outstanding service to a House, or as inducement to future loyalty. It is uncommon for a House Minor to receive such a reward, but when it does occur, that House almost inevitably finds itself elevated to the status of a House Major due to the attendant influx of wealth and prestige. Houses that are unable to rise to the occasion find themselves exiled or destroyed, perhaps just as the Emperor secretly hoped.

Directorships typically carry with them the responsibility of maintaining the most valuable trades and commodities. It is not at all uncommon for a directorship to carry with it the feudal rights to an entire planet or even a whole solar system pertinent to the production of a specific commodity. Not all precious goods are the subject of total monopoly, but it tends to be the most efficient way for CHOAM (and the Emperor) to control vital trades.

It is worth noting, however, that even the most guarded monopolies are not conducted in a vacuum. The needs for specialized equipment and materials to produce such a commodity offer opportunities for other shareholders and Houses to become inveigled in the trade. A CHOAM director will find themselves courted by numerous potential business partners, many of whom may well be hoping to uncover trade secrets and break the monopoly to their own gain.

Most directorships have been maintained hereditarily for many generations, and it is uncommon for new directors to be so invested. As such, many trades and products have been monopolized by specific Houses for many thousands of years. The most valuable of these seats find themselves the subject of generational plots by rival Houses who have nursed grudges and envy for hundreds of years. Meetings of the board are tense affairs.

THE PREEMINENCE OF MELANGE

No commodity in the known universe is more valued than the spice melange. The economic power of spice is so great as to distort the entire economy of humankind. At times, the value of a single decagram of spice has ranged as high as 620,000 solaris. Necessary as it is to the operation of the Spacing Guild's operations, and its inherent geriatric properties to which a vast number of people are addicted, melange is the most protected trade to be found anywhere.

Economic control of the spice is invariably granted to a CHOAM director at the whim of the Emperor. The Imperial House shies away from exerting direct management of spice production for two primary reasons. First and foremost, such a show of economic force would likely unsettle the balance of power between the Imperial House and the Houses of the Landsraad. Secondly, and more insidiously, the promise of control of the spice is arguably the most powerful incentive with which the Emperor may bind other Houses to him.

By necessity, control of spice production carries with it the feudal rights to the planet of Arrakis, the sole known source of melange. Arrakis is a notoriously harsh world and while the possible wealth a shareholder stands to gain from the spice trade is staggering, so too are the costs and dangers. The climate of Arrakis takes a terrible toll on both personnel and spice harvesting equipment, to say nothing of threats from the local fauna.

The harvesting and refining of melange is a complex business, and the responsible shareholder will find themselves needing to draw upon numerous outside parties, all eager to gouge exorbitant prices for their services in this critical trade. Many believe that control over spice production and distribution confers near-unlimited power, but more than one would-be master of the spice has found themselves annihilated for their efforts.

Directorate meetings are conducted on a regular schedule and attendance is mandatory. However, due to the challenges of gathering individuals from diverse star systems across the gulf of space, few directors attend every meeting and often utilize proxies to speak with their authority and vote in their places.

Meetings of the board are held on Kaitain under the protection of the Imperial House. These gatherings occur once each standard year for a general overview of CHOAM affairs and profits reporting and take place over several days. They are attended by directors and their proxies, and representatives of the Spacing Guild and the Bene Gesserit who, while silent parties in CHOAM, are sure to work in advancement of their agendas outside of official proceedings. The dinner parties and recreational lounges surrounding the events are commonly believed to be more productive for business than the actual board meetings.

Directors are invited to share concerns and suggestions at certain points and with the Emperor present these presentations usually amount to sometimes subtle but often vicious maneuvering. Any gathering of the board is a period of intense political intrigue as they essentially amount to gatherings of the leaders of the Landsraad and other great powers.

Meetings of the board may also be called to deal with crises or opportunities as they arise. These meetings are technically mandatory, but given the necessity to quickly make decisions, attendance is never total. Directors who fail to attend these meetings will rarely be formally censured but may find themselves missing important opportunities or becoming useful scapegoats.

While the Padishah Emperor is required to consult the board for major decisions, in practice his power within CHOAM is near-dictatorial and only checked by the maneuvering of the Landsraad and the influence of other powerful organizations. The Emperor is likely to act subtly through third parties in exerting his will amongst the directorate but is not above making broad and public declarations when he feels it necessary. Board members rankle at these displays of naked power but are largely reconciled to them as an unpleasant fact of life. Still, the board is an influential organization, and access to the Emperor is a prize of incredible value.

ECONOMIC REALITIES

While the above overview paints a picture of CHOAM as having relatively simple operations in truth it is a messy, often chaotic organization. CHOAM accounting is formidable, but the unimaginable number of transactions conducted each day makes it an essentially impossible task for anyone to glean anything but

a general idea of the actual sums involved. The annual reports of CHOAM profits and operations are calculated to include significant margins of error, a fact of which all shareholders are keenly aware when managing their own interests.

A certain amount of graft and embezzlement is expected as a matter of course and tolerated provided the involved parties don't blatantly show their hand. With that said, rivals will undoubtedly do their best to uncover illicit practices in the hopes of inciting censure by CHOAM authorities. Wise shareholders keep their embezzlement to a polite minimum.

This still allows ample opportunities for 'creative accounting' practices. Specialists known as *finicap* accountants with expertise in questionable book-keeping are quietly employed by any number of parties, including CHOAM itself which uses them in its auditing procedures. It is unknown whether CHOAM employs *Mentats* in these roles, but this seems likely.

Smuggling and piracy also play a role in business. It generally falls upon the shareholders themselves to deal with criminal interference in their operations. In extreme circumstances the board of directors may provide assets to increase security for a specific trade, but these costs will always be ultimately borne by the shareholder who requested them. More commonly, a House will utilize its own military and police arms in protections of its businesses or hire mercenary forces directly.

It is also not unheard of for shareholders to permit smuggling of commodities in their own control. Accepting bribes from smugglers may at times produce more illicit profit than the legitimate trade of the goods in question. Houses involved in this dubious practice must maintain strong deniability in case it comes to light.

Competition between shareholders in control of similar goods can be ruthless and at times result in the declaration of a formal War of Assassins. As a body, CHOAM takes no part in these conflicts and discourages such open violence between shareholders. Naturally, this is more obeyed in the form than the act as House wars are always good for the business of one party or another. CHOAM does offer neutral party arbitrations for disputes, though these typically occur before any blood is shed. By the time a declaration of war or *kanly* has been made, resolution of the conflict will likely fall upon the Landsraad itself.

While strife and intrigue are commonplace occurrences within CHOAM they should still be viewed as exceptions and not the rule. CHOAM has functioned and thrived for thousands of years and provides a stabilized economic base for all the great powers of the Imperium.

THE SPACING GUILD

All Houses have some means of conventional space travel, with spaceships capable of traveling from surface to space and within a star system, and armed space vessels to defend their planets from incursion. These range from lighters to great battle frigates, with a few other types between. But the means of traveling from one star system to another is beyond all but a few, and they are inevitably restricted to slower-than-light travel, which means that interstellar travel can take years, decades, or even centuries, which is an infeasible limitation.

And so it is that more than the Emperor, Landsraad, or even CHOAM, the Spacing Guild itself is the most critical entity within the Imperium. Without the ability to move goods across the universe, trade would grind to a halt and all the Houses would face economic ruin. Diplomacy would break down, and war would inevitably break out on a galactic scale, as each House could no longer depend on rapid transport of troops to keep its holdings in line. Without the means of transporting troops, the Emperor cannot hold his Sardaukar as a sword above the collective neck of the Landsraad, and the balance of power would shift irrevocably.

The Spacing Guild is quite aware of this, and acts accordingly, extracting every solari it can from those who use its services, working in tandem with CHOAM to ensure a status quo that maintains their highly profitable monopoly. For these reasons and more, the Spacing Guild is as disliked as it is relied upon, and most Houses regard dealing with it as a necessary evil. The Spacing Guild, in turn, schemes on its own to maintain its unspoken command over the Imperium and all within it, collaborating with CHOAM, the Bene Gesserit, and even individual Houses within the Landsraad, as suits its purpose. More than anything else, the Spacing Guild seeks to sustain its existence, and that means they must always have access to the spice melange, the most precious substance in the universe.

ORIGINS

The Spacing Guild is almost as old as the Imperium itself and has been inextricably tied into its inner workings since its foundation, born when foldspace technology became available. The Butlerian edicts preventing the use of thinking computers forced others to rely



SMUGGLING

Despite the Guild's exclusivity on interstellar travel, all Houses of the Landsraad have vessels capable of spaceflight, though few are capable of safe and reliable travel between solar systems due to the incredible distances and time required. In addition to these House-owned vessels, there is an inevitable criminal trade of those who wish to travel to and from space and do not wish to do so under the official auspices of the House governing that planet. Reasons vary, whether fleeing persecution, espionage, or the most basic of needs: greed.

Most planets in the Imperium have smugglers of one sort or another, and Houses deal with them in various fashions, from looking the other way, working clandestinely with them, taxing them heavily, or attempting to stamp them out. The smugglers, in turn, must deal in some fashion with the Spacing Guild, creating a case of willful blindness on behalf of the Imperium.

The Guild clearly has some arrangement with these smugglers, allowing them access to their Heighliners, likely for exorbitant fees, but their policy of strict confidentiality and refusal to let the Emperor audit their books means that people and other illicit goods move from system to system through exactly the same means as legal transport, and none but the Guild are aware of what is moving where.

As the most valuable substance in the universe, there is an entire industry devoted to smuggling spice from Arrakis and into the hands of various Houses and even the various Schools for their own stockpiling. Arrakis' spice smugglers are many and resourceful, openly defying the Houses that have held the planet, whether it be the Richese, Harkonnen, or later the Atreides. So precious is the spice that all are willing to pay dearly and risk exposure to maintain spice reserves that cannot be tracked or accounted for. The Guild, meanwhile, keeps track of the quantity of spice leaving Arrakis and who it is held by. They cannot hope to track it once it reaches the open market or the Houses that purchase it, but they are quite aware of how much spice is out there and who holds it.

In some ways, smugglers are the only members of the entire Imperium who stand outside the Landsraad and the control of the Faufreluches system. Outlaws and criminals, the ranks of smugglers are often the refuge of those on the run, and smugglers are often romanticized in ballads and stories, a refuge to any who would choose to live on their own terms. In truth, though, most smuggler bands are as regimented and disciplined as any military company, hardened folk who are driven by desperation to a life outside society.

on less reliable methods, and space travel before and without the Guild—even still—was hazardous and slow, with many more ships lost to the darkness of space or destroyed in transit.

Spice and its properties changed this fundamentally, and the Guild adapted to its use, just as its Navigators mutated to meet the needs that interstellar navigation placed on their minds, particularly the necessity of prescience. Over millennia, the Guild dominated all space travel, until the current age when its services are the sole means by which faster-than-light space travel can be accomplished. The Guild is highly proactive when dealing with any threats to its hegemony over space travel, relying on CHOAM, and if need be, the Emperor, to exert pressure on any threats to its exclusivity.

GUILD NAVIGATORS

The quality of prescience and ability to perform the complex calculations needed to plot space jumps safely require Guild Navigators, also called Steersmen, to consume massive and concentrated quantities of spice, far in excess of that consumed by any other sapient being. They eat it and must continually breathe an atmosphere of aerosolized spice gas, requiring navigators to remain in sealed chambers when on Heighliners and mobile spice tanks on the extraordinarily rare occasion when a navigator ventures outside Guild facilities.

Through processes unknown outside the Guild, Navigators are recruited from childhood and exposed to vast quantities of the spice, becoming mutants. Their eyes take on the characteristic blue-within-blue, their heads grow, features become grotesque, temples swell, their limbs extend and reposition, and their fingers and toes grow to incredible lengths, fins developing between their digits. Navigator voices change dramatically and require translators to interpret into Galach. Fully mutated Navigators are often described as fishlike in appearance, and require a constant diet of spice, swimming weightlessly on suspensor fields in mobile tanks full of spice gas. These mutations are commonplace to their kind, and Navigators become increasingly mutated as they age and advance to different stages of development. They wear formfitting garments that allow them unhindered movement within their tanks.

SPICE MUTATION AND PRESCIENCE

As described prior, until the discovery of the spice melange, faster-than-light space travel was controlled by thinking computers. After the Butlerian Jihad, such devices were forbidden, limiting space travel considerably, making it less safe and thus threatening to isolate the scattered worlds humankind had settled upon. With Tio Holtzman and Norma Cevna's invention of the Holtzman engines enabling foldspace, ships were vastly

faster, but were limited to the degree by which long-range jumps were possible. Without prescience, space travel came with the risk of utter destruction, a ship emerging into unfavorable conditions—such as the path of a star or other celestial body.

Once the psychic and mental benefits of extreme spice usage, such as prescience and enhanced computational abilities, were discovered, these qualities became essential to the Guild, which had formerly used navigational computers to calculate the distances for faster-than-light travel. With the mutations conferred by heavy exposure to spice, Guild Navigators were able to calculate safe paths, computing and predicting the near-infinite variables involved in an always-moving galaxy.

Under the influence of spice, a Guild Navigator can intuitively use prescience to plot a course suitable and safe for faster-than-light travel, allowing ships to avoid gravitational anomalies, erratic orbits, space debris, and other otherwise unforeseeable hazards to interstellar navigation. The reputation for folding space is inaccurate but adds to their mystique. Guild Navigators pilot their ships as they make these faster-than-light leaps, but they do not propel or transport the vessels themselves.

Another curious and significant side effect to Navigator prescience is that it has the effect of shielding that particular Navigator and their actions from other forms of prescience, even extending to those nearby or in close collaboration with a Navigator.

A CULT OF SECRECY

Like the Bene Gesserit, the Spacing Guild is an intensely secretive institution, and does not willingly share its methods or information about its goals and inner organization. Few outsiders have ever seen a Guild Navigator, and none have visited Guild headquarters or holdings. The inner workings of the Guild—recruiting, training, operations, administration—remain utterly opaque outside the Guild, and they strictly guard all knowledge about their organization.

The Guild's secrecy extends to even the persons of their members. Their agents often wear disguises and adopt near-anonymous, bland identities, even going so far as to cover their spice-hued blue eyes. Guild transit brokers, while visible and sometimes even working closely with Imperial Houses, are inevitably inscrutable, offering no personal information and presenting no means of blackmail or extortion.

GUILD POLITICS

Guild agents are omnipresent throughout the Imperium, operating on most civilized planets as brokers and observers. Their power is solely connected to the service they provide, though, and they are not militant on their own. The Guild has no warships, troops, or armaments. Their politics extend only to that which keeps them profitable, and thus their intervention into the Landsraad and Imperial politics are subtle for the most part, enacted through proxies. Only in the rarest or most desperate situations will the Guild venture openly into political skullduggery, and when they do so, they venture as little as possible.

As the only providers of system-to-system travel, the Guild can influence a war's outcome simply by refusing to transport frigates from one location to another or allowing more rapid transit to the side it favors. The Guild does not sell weapons, does not fight wars, but it transports arms, soldiers, and warships readily, its fees a 'soft method' of throttling the movement of arms and troops throughout the galaxy.

HEIGHLINERS

The mightiest and greatest starships in the Imperium, the Heighliners produced for the Guild are marvels of engineering, though much of their internal structure and architecture is concealed, known only to the Guild itself and those involved with their manufacture. Estimated at roughly 20 kilometers in length, Heighliners are cylindrical constructions, their immense holds capable of holding thousands of smaller ships and their contents within them, from the smallest lighters, bulky cargo transports, combat frigates, and even, on occasion, capital ships such as the Emperor's own mobile palace.

Able to encompass entire cities, Heighliners are entirely spacebound vessels, too great to land within an atmosphere. Their faster-than-light engines are capable of 'folding space' to move across the galaxy at nonrelativistic speeds, enabling rapid transport from one star system to another. Guild Heighliners are sufficiently armored to survive the rigors of this travel, and while there is no record of any Guild ships using any armaments, it is similarly unknown whether they possess any. To attack a Guild ship would be financial and political suicide, and it is possible that this is their sole means of defense. To attack the Spacing Guild is to assault the very structure of the Imperium: that should be enough to deter any aggression against them.

None outside the Guild know how many Guildsmen crew a Heighliner, though given their size the number is likely to be tens of thousands of laborers, engineers, technicians, and other specialists, all piloted by a single Navigator.

SPACE TRAVEL OUTSIDE THE GUILD

While the Guild controls all foldspace travel, it is not entirely true to say they control all interstellar travel. Some Houses have spacecraft capable of FTL travel, which they can use to cross between some close solar systems.

However, the Imperium stretches across distances that take even FTL drives decades or even centuries to traverse. This means that Guild travel is the only practical way to travel, and given its almost instantaneous nature, it is by far the fastest for even 'short' interstellar journeys. However, if a House is forced to go renegade, FTL is a good way to find a hiding place without needing to bribe the Guild.

tems, going 'dark' while in transit. The Guild links with ship communication systems and controls (and likely filters) all extra-vehicular communication. If absolutely necessary, one House's ship can speak to another, but the Guild is monitoring all of this communication.

Ships traveling on Guild Heighliners must submit manifests of all cargo they carry, as well as the identities of all personnel being transported, giving them unrestricted insight into the affairs and trade practices of all those they serve. The Guild not only knows what goods are being transported, they know who is going where, giving them a clear picture of any House's status and activities.

Finally, and most significantly, vessels within a Guild transport are absolutely prohibited from interacting with one another, in any fashion. The Guild's greatest strength is the guarantee of security while in transit, so Houses who are mortal enemies may travel parked next to one another within a Guild vessel and expect utter safety. In some cases, the Guild may even be transporting ships on opposite sides of a conflict to the site of their future engagement.

The penalty for disobeying any of these Guild restrictions is inevitably lack of access to the Guild's services, whether short-term or for longer, depending on the severity of the infraction. Without space travel or interstellar communication, a House is cut off from the Landsraad and the rest of the Imperium, deprived of trade and ultimately, of spice.

SPACE TRAVEL

Traveling on a Guild Heighliner is thus a relatively casual experience. Other than the brief vantage of seeing ships entering the Heighliner's hold, there is little to see. Once the Guild has connected to a House's ship's communication and sensor systems, each ship is self-contained. Passengers are free to do whatever they choose to while waiting.

There is no sense of velocity, no shift in gravity, no appreciable clues that the Heighliners are traveling at faster-than-light speed. Guild Heighliners make many stops as they gather passengers from adjacent star systems. After an interval—which could be minutes, hours, or even days—the Guild contacts the ships it has transported and informs them that they have reached their destination and can disembark and land at their own convenience. The amount of time one spends on a Guild Heighliner has little to do with the amount of time it takes to actually fold space, which is near-instantaneous.

It is entirely possible for a Major House, or perhaps the Emperor himself, to commission transport alone, but such an expense is beyond the means of most and done only in the most extraordinary of circumstances. In these cases, it is highly likely that the Guild will only agree if the reason for this transit aligns with their own purposes.

GUILD POLICIES

As the exclusive provider of system-to-system space transport, the Guild can, and does, impose drastic policies and restrictions upon those who they convey. All agree with these conditions because the alternative is neither desirable nor acceptable.

The first edict is that passengers on Guild vessels are absolutely forbidden to leave their own ships, save for the most extraordinary of circumstances, and only when so explicitly permitted by the Guild. Trips outside one's vessel are limited to conveyance via a Guild shuttle to a Guild-controlled facility, resembling a lounge or transit hotel, where negotiations or communication may be held. Guild personnel staffing these facilities are always unmutated humans, specifically chosen for their blandness of appearance and affect.

Ships within a Guild Heighliner or other Guild transport are expected to shut down their sensors and any surveillance capabilities, as well as their communications sys-

THE ORDER OF THE MENTATS

What is a Mentat? A Mentat is one of 'a class of Imperial citizens trained for supreme accomplishments of logic and deduction.' They are essentially human computers that apply their ethics in a neutral manner along with unbiased logic to solve problems and formulate plans. Unlike their machine counterparts, their memories and perceptions enable them to perform supra-logical hypothesizing.

Mentats can process immense amounts of data from various sources by tapping into 'the naive mind'; a mind without preconceptions, analyzing data with extreme precision and speed. They notice logical patterns to deliver predictive outcomes based on analysis of data and potential influencing factors.

Being a Mentat requires a specific mental aptitude and mental conditioning that is rare in most people. No two Mentats are equal, as each person is unique and cannot operate at the same capacity. As universal demand is high, most Mentats easily find employment regardless of their status.

HISTORY OF THE MENTATS

The Order of the Mentats was birthed from the Butlerian Jihad. The century-long crusade ended with the destruction of all thinking machines and a decree that no machine can be built in a similar fashion to a human mind. That edict destroyed technological and social structure of the time and created a massive power vacuum in the universe.

THE FIRST MENTAT

The first Mentat came into being as the result of a wager between Erasmus and Omnium, regarding humanity's usability. This altered the course of the universe. Omnium bet that a human could not be improved into a logical being, regardless of Erasmus' efforts. Erasmus accepted the wager, and Omnium selected the wildest looking child it could, an enslaved person on the planet of Corrin. Erasmus chose a regal sounding name for the child, Gilbertus Albans, so that others would hear the nobility in it and respond accordingly.

Erasmus encountered countless failures before finding an approach Gilbertus would respond to. Eventually, the boy began displaying the first human computer-mirroring cognitive and calculation abilities that rivaled thinking machines. With that step established, Erasmus began teaching Gilbertus how to organize his thoughts and think logically as a computer would. This intensive training, higher recall, and conditioning allowed Gilbertus to remember everything. The man processed information at lightning speed, reciting observations, computer level calculations, and maintaining numerous tangent conversations with ease. The two formed a familial bond that inspired the independent robot.

Erasmus did everything possible to make Gilbertus the pinnacle of what a human could be. The first step was a procedure that extended the human's life span, done illegally and without Omnium's agreement. Next, Erasmus instituted an intensive physical exercise routine. Erasmus' fondness for Gilbertus led to the thinking machine creating the honorific title of 'Mentat', bestowed on Gilbertus due to his enhanced intelligence.

Gilbertus risked his own life thwarting Erasmus' plans to kill millions of humans. The thinking machine's affections for the human caused it to abort its plan. This was one of the key events that aided the Butlerian Jihad in destroying the AIs. As the last of the thinking machines were destroyed, Gilbertus rescued Erasmus' memory core and spirited it away with him.

BUILDING THE ORDER

After 70 years of training, Gilbertus opened a school on the planet Lampadas to teach what he had learned. The school was also where he protected the memory core of his friend and mentor Erasmus, a thinking machine. The new Mentat order walked a balance of either conforming to the Jihad or denying them.

This nascent order knew it needed to become part of the Empire to avoid falling to the wayside, so they made a deal with CHOAM agreeing that the Mentats would use their considerable intellectual and predictive capabilities in service of the Empire, in exchange for employment and protection.

MENTAT TRAINING

MENTAT DISCOVERY

A prevalent rumor has followed the Order of Mentats since the time of Gilbertus that babies are taken from their homes to begin learning the art. The younger the child, the more likely they are to succeed. While it has been lucrative for the Mentat Order to remain silent on the matter, it is, in fact, false. The ability to become a Mentat is rare and only a fraction of a fraction of a percent possess the capability to do so. That ability can be harnessed and trained at any age.

Rumor has it that every Major House and anyone of sufficient means has their children undergo an assessment by a senior Mentat teacher, a test to determine their suitability. If found acceptable, the parents are offered the following choice:

- First, allowing the child to formally join the Mentat Order, leaving with the teacher.
- Second, requesting a Mentat advisor to train the child if they do not already have one in their household.
- Third, refusing for the time being.

THE JUICE OF SAPHO

The juice of Sapho is derived from a plant found on the planet Ecaz. Once harvested, it is made into a liquid high energy drink, a stimulant with the unparalleled ability to temporarily amplify a Mentat's mental abilities. Extended use of the juice is addictive and obvious, leaving deep ruby stains on one's mouth and lips.



MENTAT TRAINING FOR NON-MENTATS

Mentat training is intense and essential for the development of mental abilities. Anyone receiving this training gains a better insight into their universe, and it can be used to improve their cognitive abilities to some small extent. For instance, at the Emperor's request, Gilbertus Albans took in Anna Corrino to do Mentat training to improve her focus.

TODDLER

The Order of Mentats prefers to start training recruits from childhood. Research and millennia of experience have shown that children able to bond with their parental structure are able to relate better with non-Mentats after becoming Mentats. It is believed that those initial developmental years establish a baseline personality and loyalty that is built upon. While Mentats approach complex issues from a more neutral standpoint, it is critical for the acceptance of the outcome that they relate to their employer. The reason for their training is kept secret from the students, whether in a school or learning under a single instructor.

It is rumored that Bene Tleilax have abducted newborns to test if the transition to a corrupted, or 'twisted', Mentat is less complicated with a younger subject. To date, no records of the outcome have been discovered. See **Twisted Mentats** (p.61) for more information.

The curriculum is identical for the first handful of years, whether the young Mentat-in-training is taken to an official academy or trained by a private advisor. Training at this stage resembles training any student of their age would take, with a gradual increase in the difficulty of tasks on a weekly basis. The taskings are regulated per student, with more gifted students receiving more computations until they are increasing at a percentage rate suitable for their group.

The course of study focuses on expanding the storage capacity of each student's mind through presenting increasing amounts of data for them to retain and repeat back to their instructors. The first object is accuracy of the data, then volume of data retained, and lastly how quickly the data is repeated back to the instructor. The instructor encourages physical activity and provides time for the students to do so. By the end of this early training, their instructor can assess the capabilities of all the students to a specific degree of accuracy.

ADOLESCENCE

The next stage involves more physical demands on the young Mentats, placing them in hostile, dangerous, and extreme conditions. By this age, the students excel at the foundations of being Mentats, and the next step in their training is to implement different stresses and dangers they may encounter, during which they are expected to remain calm.

This level of training is intense and generally lasts for a year or two at most. The foundation of the first stage allows for an accelerated second stage. All future Mentats are taught the basics of combat, and their ability to read the slightest shift in body movement makes them exceptional combatants.

THE CHOICE

Upon reaching a specific threshold in their development, every Mentat-in-training is told the truth about their training. Some have deduced what has been transpiring and others remain unaware. They are provided the choice to continue their training or to decline and go back to their daily life. The Mentats that decline go into the universe with their new training and discipline, frequently finding work for Minor Houses that cannot afford a full Mentat advisor.

BECOMING A MENTAT

The students who decide to continue their training receive the final years of Mentat conditioning, which forever changes them. This turns them from highly-gifted students into true Mentats, worthy of the name and capable of handling all the tasks required of them. Many pupils never proceed beyond this stage, and it is rare that anyone is trained to this degree outside of a Mentat training facility.

To increase their marketability, each Mentat is taught a specialty in demand at the time. The highest-scoring student of each class can choose the specialty they desire, creating an air of competition within the school. The three most frequently requested specialties are assassination, strategy, and archives.

TWISTED MENTATS

Originally created by the Bene Tleilax, twisted Mentats possess all the ability of other Mentats but operate without any moral or ethical constraints. They are essentially brilliant sociopaths, with little concern for human life. These corrupted Mentats are highly prized by the more questionable noble Houses, as they provide an additional resource denied to nearly every ethical House.

There is officially no known school that creates twisted Mentats. Inquiries to the Order of Mentats regarding them garners only silence, a refusal of any services, and possibly being reported to the Emperor.

MENTATS AND THE KNOWN UNIVERSE

Generations ago, the Order of Mentats established a deal with CHOAM to provide employment for their members and aid the empire. The employment varies from archivist to advisor to strategists in the court of Emperor Shaddam IV. Many find themselves in service to various Houses as political advisors, military counsels, sanctioned teachers for younger Mentats, or to train future leaders in the fundamentals of Mentats to increase their own focus.

The Bene Gesserit themselves have a great need for Mentat training, and their own Mentats are frequently pre-conditioned Reverend Mothers.





THE BENE GESSERIT ORGANIZATIONAL HIERARCHY

The Bene Gesserit have a strict power structure, with the Mother Superior in command.

MOTHER SUPERIOR

She oversees the entire Sisterhood. Though she relies on the counsel of select Reverend Mothers, she is the highest authority in the organization. Through Other Memory, she has incredible wisdom and insight; however, past Reverend Mothers sometimes withhold information even from this formidable woman, splitting the responsibilities of the Sisterhood between her and the Kwisatz Mother.

REVEREND MOTHER

A Reverend Mother is a member of the Bene Gesserit who survive 'the agony' (see p.57), an ancient Bene Gesserit ritual in which the candidate is poisoned and forced to metabolize and neutralize that poison internally, unlocking hidden power and enlightenment. Those who succeed gain access to Other Memory and the respected title of Reverend Mother. Those who fail, die painfully.

KWISATZ MOTHER

The Kwisatz Mother is a Reverend Mother, though hers is considered a 'hidden rank' and kept secret even within the Sisterhood. Many other Reverend Mothers in the organization don't even know why she has such a high standing within the organization. This particular Reverend Mother has special insights into the ultimate goal of the Bene Gesserit Sisterhood: the creation of the Kwisatz Haderach through genetic manipulation performed through breeding. This powerful woman has an even deeper connection to Other Memory than the Mother Superior does.

TRUTHSAYER

Some Reverend Mothers can enter a truthtrance, giving them the ability to detect lies and falsehoods. Traditionally, the Padishah Emperors have kept a Truthsayer near at hand to keep their people honest and help them root out any kind of insubordination... having a window into the inner workings of the Imperium from a well-embedded, loyal Reverend Mother source is also valuable to the Bene Gesserit.

SISTER

These women have completed their Bene Gesserit training but have not undertaken the ritual transformation into becoming a Reverend Mother. They still have **prana-bindu** control over their bodies (described below) and many of the other seemingly magical gifts that the rest of the Imperium both fears and admires, however. It is not unusual for sisters to act as the hands of the Bene Gesserit throughout the Imperium by becoming spies, assassins, advisors, wives, concubines, and agents furthering the goals of the order.

INITIATE

Still in training, initiates are not yet full-fledged sisters. Some women, particularly those whose families pay to have them educated by the Bene Gesserit, will never become a Sister and will never be made privy to many of the order's secrets. However, even these novices know more than they let on. They may appear fresh-faced and inexperienced, but these young members of the Sisterhood are not to be underestimated.

THE BENE GESSERIT

This mysterious Sisterhood is one of the most ancient schools of the Imperium. On the damp, inhospitable planet of Wallach IX (see p.83), the Bene Gesserit Mother Superior guides humanity with a subtle hand from the massive Mother School complex. Women from around the Imperium go there to endure the physical and mental conditioning the Bene Gesserit are famous for. So renowned are they, that even the Houses Major and Minor pay to have their daughters educated there. In the school complex, initiates learn control, discipline, and loyalty to the Sisterhood.

HISTORY OF THE BENE GESSERIT SISTERHOOD

Almost 20 years after the Omnium Scourge's initial wave of destruction, a mutated version of the deadly retrovirus appeared again on the planet Rossak. This new version of the plague, which was later called the Rossak Epidemic, had an even higher mortality rate than the original. Raquella Berto-Anirul and Dr. Mohandas Suk, future founder of the Suk School, were dispatched to the planet by the Humanities Medical Commission (HuMed).

On the surface of Rossak, Berto-Anirul treated patients and searched for an effective remedy. However, she faced hostility and resistance from the local population and their leaders, the Sorceresses of Rossak (following). Eventually, she contracted the new retrovirus herself and almost succumbed to the devastating disease. Her friend, Jimmak Tero, treated her with healing waters from a pool deep in the Rossak jungles that only he and a few other Rossak outcasts knew about.

Some combination of the flora-infused water, the retrovirus that raged through her body, and the harsh environment of Rossak itself changed Berto-Anirul on a cellular level. When one of the Sorceresses tried to poison the doctor, she realized that she could transform the toxin into something harmless with her own biochemistry, allowing her to survive the assassination attempt.

Not long after, Berto-Anirul went on to lead the Sorceresses and transformed their organization into the Bene Gesserit Sisterhood famous across the Imperium today.

AGENDA

The Bene Gesserit have several goals, though none are openly discussed with those outside the Sisterhood.

- ✧ This arcane organization has a plan for all of humanity. With gentle coaxing, political alliances, social maneuvering, the occasional veiled threat, and the rare assassination, the Sisterhood focuses the Imperium on the direction of its choosing and toward the future they believe is the correct one.

- ✧ They ensure their influence and income by making their schooling highly desirable. For a fee, they instruct daughters from noble Houses across the Imperium.
- ✧ Their most secret goal is to create the Kwisatz Haderach, a being who could acquire knowledge even beyond the Reverend Mothers with their Other Memory and connect or manipulate space and time. In pursuing this agenda, they have manipulated several bloodlines in secret for centuries.

Their unique combination of communication ability, information-gathering expertise, and emotional manipulation makes them formidable spies, and the Sisterhood maintains its own spy network that extends to nearly every House within the Imperium, with few exceptions. Even those Houses that do not have a Bene Gesserit-trained member as a counselor, spouse, or concubine, may have one or more operatives working clandestinely within their households and reporting to the Sisterhood.

THE MISSIONARIA PROTECTIVA

Within the Sisterhood is a long-established group called the Missionaria Protectiva dedicated to spreading the **Panoplia Propheticus** across different worlds. Composed of fables, superstitions, and myths, the Panoplia Propheticus has an appealing message that lends itself to integration with local doctrines and religions. These legends seed beliefs about powerful women and various prophecies.

THE SORCERESSES OF ROSSAK

On the hostile jungle planet of Rossak, a matriarchal society of telekinetic sorceresses reigned even before the Butlerian Jihad. A Supreme Sorceress led the community. They kept detailed genetic records. The Bene Gesserit acquired this archive after Berto-Anirul took over the Sorceresses of Rossak and transitioned the group into the Bene Gesserit Sisterhood.

Depending on the living conditions of a planet, the current culture, and a variety of other considerations, the Missionaria Protectiva might choose different narratives or versions of the Panoplia Propheticus to circulate.

After the Missionaria Protectiva completes its work on a planet, any Bene Gesserit can identify particular phrases and imagery from the collection of the Panoplia Propheticus even millennia later. These recognizable features provide a known religion or belief that a Sister can tap into for support thanks to her understanding of the Panoplia Propheticus. By using the myths and superstitions dispersed throughout a planet by the Missionaria Protectiva, a Sister can make herself seem like a legend come to life, allowing her to work with or even exploit a local population.

BENE GESSERIT TRAINING

Members of the Sisterhood go through extreme instruction and education to learn how to control their own bodies and the bodies of those around them. Mastery of these skills may appear supernatural or superhuman but are the result of diligent practice from the learner. Not all can withstand the harsh mental and physical conditioning the Bene Gesserit put their students through.

PRANA-BINDU

As they train, Bene Gesserit work to discipline their minds and assert control over every part of their bodies. This *prana-bindu* conditioning encompasses a body's musculature and nervous systems, giving them the ability to isolate the movements of individual muscles. They can use this command over their own bodies to stop their hearts, appear dead by slowing their heartbeats or breathing, strike an opponent with uncanny precision, work on complex technology that requires supreme dexterity, or even perform delicate surgery that needs an inhuman level of precision.

The Sisterhood uses *prana-bindu* control in their unique form of martial arts. Called the "weirding way" by the Fremens (p.71), this hand-to-hand combat style uses *prana-bindu* training to a devastating level. An experienced member of the Sisterhood can fight with almost superhuman speed and movements. With such minute control over every muscle, only the Emperor's Sardaukar rival their deadliness.

VOICE

Because of their incredible physical prowess, members of the Bene Gesserit can also take advantage of Voice. By modulating the timber, pitch, and cadence of their words, a trained member of the Bene Gesserit can compel a listener to follow her commands. A vulnerable individual subjected to Voice will follow almost any orders, from physical actions to changing their very thoughts. Strong Reverend Mothers, the deaf, and iron-

willed Mentats who understand how Voice works, are some of the very few who can resist the compulsion.

Voice is a dangerous tool in the wrong hands. A Bene Gesserit can cause a person to hurt themselves or someone else. They can also pry codes or information from an unwilling mind. However, use of Voice can also save a life. Instructing someone to seek help or put down a weapon can defuse a critical situation.

INTERNAL BIOCHEMISTRY CONTROL

In addition to their muscular and nervous systems, Bene Gesserit sisters and Reverend Mothers have incredible command over their internal biochemistry, something they keep a secret from the Imperium at large. Using their own bodies, they can change the chemical composition of a poison, removing its toxicity. They can also destroy diseases introduced to their bodies or even make a potential affliction go dormant, reawakening it to infect another individual at will. This biochemical control gives a Bene Gesserit the ability to create antidotes or even cures from her own internal systems. It also means she can manufacture poisons, as well. Attempting to drug, administer poison to, or infect a member of the Sisterhood is ill-advised.

HEIGHTENED OBSERVATION

The Bene Gesserit order also puts an emphasis on detailed observation. The tiniest flick of the eye or hesitation in speech will give a Sister a mountain of knowledge about whether someone is telling the truth, hiding something, or has a separate agenda (although only a Truthsayer knows with absolute certainty). The way a jacket folds might indicate a concealed weapon. Favoring one leg might hint at an old injury and a potential weakness to exploit. Bene Gesserit are trained to be perceptive and insightful. This training overlaps considerably with that of the Mentats and is often instilled through a similar process to Mentat methodology.

SPYING AND SOCIAL MANIPULATION

Perhaps the greatest tool in their arsenal of abilities is their extensive conditioning in the arts of interpersonal communication: whether conversation, oratory, negotiation, diplomacy, and even seduction. Bene Gesserit Sisters are adept at detecting and following verbal and nonverbal cues, directing communication to serve their goals, such as learning more about a subject than they suspect, implanting ideas or emotional responses subconsciously towards a favorable result. Furthermore, this extends to human intimacy: conditioning subjects into 'sexual imprinting' where their sexual gratification is explicitly tied to that Bene Gesserit, heightened even to the degree of addiction.

These, and other subtler methods, make their members consummate spies and political operatives, achieving the ends of the Sisterhood while exerting little overt or direct influence. This is a long-suspected and mostly-accepted consequence to sharing one's bed with a Bene



Gesserit-trained spouse or concubine: that they may be subtly influencing their partner to view the Sisterhood favorably and to take actions that serve their own agenda, even if subtly and unconsciously.

HIDDEN MEANS OF COMMUNICATIONS

Additionally, throughout their education, members of the Sisterhood learn a variety of ways to communicate with each other. Key phrases, such as “on that path lies danger,” alerts a Bene Gesserit to look for a secret message from another member of the Sisterhood. They may use hand signals or coded dots (p.208) to give each other information they do not want to provide to outsiders. Communication between Sisters may even be centuries removed from each other as in the unique terms or imagery used in the Panoplia Propheticus (p.55) on different planets.

THE TRUTHSAYER’S TRUTHTRANCE

Some extraordinary Reverend Mothers can enter a truthtrance, allowing them to identify when someone is lying. This ability, both celebrated and feared throughout the Imperium, is sought after by those looking to negotiate, solidify trade agreements, or even interrogate a prisoner. The Padishah Emperor always has a Bene Gesserit Truthsayer by his side. While most Truthsayers need a

special Truthsayer drug (p.208) to enter a truthtrance, a few exceptional Reverend Mothers can choose to go into this hypnotic state at will.

THE AGONY

The agony is an old tradition from the first days of the Bene Gesserit. Raquella Berto-Anirul was the first to survive and gain all the benefits of the agony. In this trial, a Reverend Mother candidate imbibes a fatal poison and must use her control over her own body chemistry to alter the toxic chemical composition into something harmless.

If a Sister manages to neutralize the poison and lives, she becomes a Reverend Mother. Reverend Mothers become receptive to genetic memory. This is known as Other Memory, a collection of the personalities and memories of previous Reverend Mothers. This Other Memory can provide wisdom, advice, and guidance to a Reverend Mother as though speaking through the echoes of the past. It is only available when one Reverend Mother is on hand to bestow this gift, however.

Other Memory can pick and choose what to reveal to specific Reverend Mothers, however. The Kwisatz Mother (p.54) is a special case and has more interaction with the past Reverend Mothers in the shared Other Memory than any other living Reverend Mother, even the Mother Superior.

REACH ACROSS THE IMPERIUM

This ancient Sisterhood has placed spies and agents throughout the Imperium. One can find Sisters throughout the families of the Landsraad, within the Padishah Emperor's own House, and even working as laborers where it can help them further their agenda. There are few places a Bene Gesserit cannot infiltrate. However, few, if any, have made it to the mysterious planet Salusa Secundus to monitor the Sardaukar-in-training. Other highly insular cultures like the Tleilaxu or the Fremens make it unlikely that one will find a member of the Sisterhood in their midst... but not impossible.

ATTITUDES TOWARD THE SISTERHOOD

Because of the noble and distinguished reputation of their education, many families of the aristocracy send their daughters, wives, and other women in their lives to the Bene Gesserit for instruction. This perpetuates the cycle of keeping the families of the Houses Major and Minor invested in the organization... and the influential women of the Imperium loyal to the Sisterhood.

With their far reach across the Known Universe, many different factions are willing to ally themselves with the Bene Gesserit. They have a stake in CHOAM, for instance. Even the Padishah Emperor Shaddam IV married a Bene Gesserit of Hidden Rank.

Despite their significance within the Imperium, many do not trust the Bene Gesserit or actively disapprove of

their political and social machinations. The Sisterhood has been known to blackmail those in power to reach their inscrutable goals.

With their highly visible presence among the Houses and their persistent, though unobtrusive, participation in all the major centers of power, those who are also vying for dominance in the Imperium often mistrust the Bene Gesserit, even if they outwardly ally with them. The Bene Gesserit Sisterhood's secretive withholding of their ultimate goals or aims does not help their reputation with those who are already wary of them.

Beyond their political and social dealings, the rumors and suspicions about their abilities give the Bene Gesserit a certain mystique. Though the Sisterhood keeps many of their powers, like their control over their internal body chemistry, secret, some are well known. For example, the Padishah Emperor always has a Truthsayer on hand. The unusual, mysterious abilities of the Bene Gesserit are the result of intense training but seem like magic to the layperson.

Between their discreet, evasive movements within the political and social circles of the Imperium, their knowledge of the business of the Imperium, and their incredible abilities, many view the Bene Gesserit as witches. Whether that has negative or positive connotations depends on the individual, their circumstances, and sometimes the individual Bene Gesserit involved.

The Bene Gesserit, themselves, seem to encourage all aspects of their reputation—positive, negative, supernatural, and esteemed. The more tools at their disposal, the more options they possess to guide the Imperium in the direction they believe is the right one.

THE LITANY AGAINST FEAR

I must not fear.

Fear is the mind-killer.

Fear is the little-death that brings total obliteration.

I will face my fear.

I will permit it to pass over me and through me.

And when it has gone past I will turn the inner eye to see its path.

Where the fear has gone there will be nothing.

Only I will remain.

THE SUK MEDICAL SCHOOL

Among all the powers and factions found in the Imperium, none are as universally respected as the graduates of the Suk Medical School. When the great and powerful seek the hands of a doctor it is to the Suk that they go.

HISTORY

The Suk School was founded by the legendary Dr. Mohandas Suk soon after the end of the Butlerian Jihad. Dr. Suk was instrumental not just in humanity's efforts in that war, but to its very survival against the thinking machines. The collection of doctors, researchers, and scientists he assembled has arguably never been rivaled and it was this group that formed the core of the Suk School in the aftermath of the war and the signing of the Great Convention. This private institution became the premier center for study of medicine in the Known Universe.

Doctor Suk's principles guiding the advancement of the human mind through mastery of the human body are very much present in the school's teachings, and his mandate to do no harm to another human being guides every graduate. Though admission is costly, the Suk School has continued to train medical practitioners and provide a place of research and collegial review for millennia.

ADMISSION

The school provides education and training for thousands of roles from simple medical technicians to the master physicians that attend the nobility of the great Houses. Admission is theoretically open to all. However, enrolling in even the school's most rudimentary courses is a great expense and competition for entry is fierce. As a result, admittance is largely afforded to those who not only meet the high standards of intelligence required by the school, but those who can also afford the fees. The school does make some effort to seek out promising individuals who receive their education pro bono, but these students will find themselves indebted to the school for decades, if not the rest of their lives. All graduates of the institution are required to remit a portion of their fees to the school for the duration of their careers.

Once accepted, students find themselves undergoing an intense education not just in practical medical matters but in the liberal arts, as well. The Suk School is no mere trade school from which workers are churned out, but an institution that produces practically trained but critically minded thinkers. Even the most modestly trained medical technician can declaim the core philosophical rhetoric of the school and defend it if need be.

Technicians and medical aides may spend up to a decade in their training before being certified in their field of study and graduated. These individuals most

commonly return to their patrons and serve in their intended capacity, but some may find employment through the school's extensive network of connections.

Students studying to become full Suk doctors spend a similar length of time in course work and being tested before completing their training in the Inner School. Those incapable of passing the entrance exams will need to find employment as highly trained medical orderlies and common doctors.

THE INNER SCHOOL

Admittance to the Inner School is possible only for those students able to pass rigorous testing in both medical knowledge and philosophical matters and willing to shoulder the astronomical costs of such an advanced education. Those who make the cut find themselves subject to years more education in practical medicine and enduring the infamous Imperial conditioning.

Suk doctoral candidates must master a staggering body of information and its practical application. They are trained in a comprehensive knowledge of human anatomy and pathology and can make complex and accurate diagnoses with only minimal aid from advanced instruments. They are also trained in numerous aspects of chemistry and molecular biology, enabling them to synthesize custom medications to meet the needs of every specific patient. The regimen of coursework is interrupted only by thousands of hours assisting instructors in surgery and practical medicine.

The expertise gained from this training cannot be overstated. A graduate of the Inner School possesses one of the most superlative medical educations in all of history. A Suk doctor can make insightful diagnoses even in challenging circumstances, perform emergency surgery, manage a patient's pain, and do it all with the assurance of a seasoned expert.

CONDITIONING

All graduates of the Suk Inner School are subjected to the so-called 'Imperial' conditioning; an unbreakable psychological inhibition against causing harm to a patient. Developed in part by Doctor Suk himself, the Imperial conditioning is the fulfillment of his wish that his students and protégés would act only in the betterment of humanity.

This conditioning takes years to bestow upon a candidate and is the result of intense guided manipulation by the school's specialists. The philosophical education students mastered as undergraduates serves as a strong framework for the instilling of the conditioning. Inner School students continue to receive education in philosophy with an increasing emphasis on rhetorical study. These challenging philosophical declaiming and



defense sessions are augmented by increasingly intense hypnotic and hypnogogic therapies.

Students are placed in a receptive psychological state through a combination of *prana-bindu* muscular manipulation and neurochemical drugging. Once in this state, the student will be subject to lengthy rhetorical interrogation and subliminal direction. These sessions can last more than 36 hours and utilize exhaustion to further condition the subject. During their final years of study in the Inner School a doctoral candidate will be subjected to conditioning sessions even in their sleep. It is not unheard of for students to die during the process.

A candidate who survives Suk conditioning will find themselves unable to knowingly cause malicious harm to another human being. The conditioning has never been known to be broken and is of such potency that even the emperors have entrusted their care to Suk doctors, hence the commonly used term of Imperial conditioning.

DROPOUTS

While most of those admitted to the Inner School do go on to graduate, a small number are psychologically broken by the demanding and relentless training. These individuals usually find employment as proctors or low-level instructors for the school itself, but some few do end up practicing medicine in the wider universe. Even a failed student from the Inner School possesses a knowledge and skillset of great value.

These doctors are tattooed to mark them as graduates of the Inner School. The tattoo is in the form of a simple black diamond upon the forehead. Under simple molecular observation this tattoo bears certain irreproducible characteristics, ensuring the veracity of an individual's identity as a Suk doctor.

SUK DOCTORS

It is only upon the completion of the conditioning that a student is allowed to graduate. A graduate of the Inner School is also permitted to wear their hair in a ponytail bound with a particular silver ring.

Suk doctors are commonly employed by all the great powers of the Landsraad as personal physicians, but also as medical overseers for armies, directors of medicine for planetary governors, and as lead researchers in medical industries. Some Suk doctors choose—or are invited—to serve the Suk School itself as researchers and instructors.

Suk doctors are not cheap, and their services come at an exorbitant price, a portion of which is always remitted back to the Suk School as a condition of their graduation. Suk doctors are forbidden to act without payment, and for this reason most Houses of means retain a Suk doctor permanently, choosing to pay the higher rates than to risk a delay in treatment while the doctor's fees are negotiated or arranged.

No matter their ultimate vocation, a Suk doctor is in possession of a formidable mind, a skillset of incalculable value, and the trust of the most powerful people in the universe.

THE BENE TLEILAX

While the sisters of the Bene Gesserit have a reputation as 'witches', there is another organization more reviled but whose services are just as in demand. The Bene Tleilax, often referred to as the 'dirty Tleilaxu', are biologists first and foremost. They offer a variety of products to the Major Houses, many of which are humans genetically crafted for a particular purpose. To say their methods and scientific morals are questionable is an understatement. However, very few people are privy to the nature of their technology as they are an extremely isolationist society. Their home planet Tleilax is ruled by a totalitarian theocracy, dedicated to the strictures of the Butlerian Jihad, and outsiders are not welcome.

While Tleilax is a planet of secrets, they are not mysteries that anyone wants to uncover. Tleilaxu products are used all over the Imperium by all the Houses. But most people find the nature of these products unsettling, to say the least, and know that if they understood more about how they were made they would be sickened to the core. So, their customers are content to not ask questions, buy their products anyway, and denigrate them to assuage their guilt. The only thing anyone really knows is that most of their technology is grown using 'axlotl tanks', although the nature of these tanks remains a mystery. It is generally believed that uncovering the secrets of these tanks is the key to understanding all the Tleilaxu technology.

Officially, the Tleilaxu take no part in Imperial politics. They insist they are simple scientists, only interested in expanding their knowledge. They only sell their biological wares so they might subsist and fund their research. Otherwise they have no interest in the dealings of the Imperium and its people. This is, of course, a lie. Like any other agency, they have spies and pawns in all corners of the Imperium looking to advance their agenda.

While they do not openly advise the Major Houses, sit among the Landsraad as the masters of a planetary domain, or work in direct service of the Emperor like a Truthsayer, they meddle as much as the Bene Gesserit. While their precise agenda is unclear, they broadly seek to control the Houses with the products they sell them. Their agents seek out the secrets of these Houses—their weaknesses and desires—so they might sell them exactly what they want. Something so essential that their customers will do anything; agree to anything, to get more. In this way they hope to one day make puppets of the entire Landsraad.

The biological products available from the Tleilaxu are myriad and usually bespoke. You tell them what you want made and they make it, from guard dogs to augmented agents to slave workers who will never revolt. The Tleilaxu will provide. However, they are particularly renowned for the following products.

FACE DANCERS

Originally designed as entertainers (a service they still perform) **Face Dancers** also happen to be the perfect spies. They are certainly used by the Tleilaxu as their primary agents. The main skill of a Face Dancer is the ability to change their body and features to mimic anyone, male or female, fat or thin, even tall or short, to an intimate level.

While the physical ability is imparted in their creation, it is their training that makes them truly deadly. They are gifted with *prana-bindu* skills worthy of a Bene Gesserit. This allows them the control over their muscles and physical augmentations that they need to transform along with frighteningly fast reflexes. Coupled with an eye for detail, a Face Dancer can mimic the gestures and mannerisms of whatever disguise they take on. This not only makes them incredible spies, but equally deadly assassins.

Thankfully, those skilled in hyperawareness can often spot a Face Dancer, but most people wouldn't notice if their own mother was replaced by one.

TWISTED MENTATS

Another 'people product' in secret demand by many Houses are the specialized 'twisted' Mentats produced by the Bene Tleilax. Standard Mentat training makes most Mentats moral and truthful people. They learn that bad or corrupt data produces bad calculations. While most Houses want someone honest and loyal, other nobles want a Mentat that can lie and cheat so they might create more devious plots. Generations ago, the master genetic manipulators, Bene Tleilax, discovered a way to train twisted Mentats at their own Mentat training academy. The discovery is kept a secret to all, with only a few knowing of the possibilities. One of these few is Baron Vladimir Harkonnen, thrilled and willing to pay an exorbitant amount to have the twisted Mentat Piter de Vries in his service.

The Tleilaxu are adept at manipulating the personality of the Mentats they produce so they can cope with the contradictions in the corrupt training they give them. This often produces Mentats who are skilled assassins.

The training regimen for twisted Mentats involves the Bene Tleilax breaking the Mentat's mind down and rebuilding them in whatever fashion their customer requested. The exact process of this was different per student and request. This may have required life-and-death encounters, extreme experiences from pure pleasure at one extreme to torture at the other. By the end of their training, the twisted Mentat is apparently loyal to their employer, their minds malleable to the wishes of the Bene Tleilax. They regularly have secret commands

put in place by the Bene Tleilax and are often more loyal to their makers than to their actual employers.

A problem with twisted Mentats is that the adaptations to their personality often give them some deep seated psychological issues. Many are psychopaths, although that is often just what the Tleilaxu's clients ordered.

These Mentats are also described on p.53.

MLEILAXU EYES

Replacement limbs and organs are also available from the Tleilaxu. While such prosthetics are not always biological in nature, they are always biologically compatible. While there are other prosthetics on the market, only the Tleilaxu can create more delicate replacements, like eyes. Tleilaxu metal eyes are one of their most famous advances, and fully restore vision. While they are a little disconcerting to others, the eyes work just as well as natural ones, although they lack any expression. It is hard to mistake them for natural eyes, and some people would rather be blind than marked by the Tleilaxu in some way.

Otherwise a variety of Tleilaxu prosthetics are available. However, their research has not developed as far as it might in this area. When making technological devices that connect to a person's nervous system, they are beginning to take a step towards machines that think. After all, if such technology is linked to the brain and nerves, what part of the person is doing the thinking? It is a line even the Tleilaxu will not cross, at least not openly.

GHOLAS

Perhaps the greatest and worst thing the Tleilaxu can do is bring people back from the dead as gholas. Publicly, little is known about the process by which they take a dead body to their axlotl tanks, apparently restarting its biological processes and making dead flesh live again. This means that it is usually only done when people die young or from battle wounds. The body will continue to age, and at a certain point the system is just too old and degraded to function. The process usually destroys some of the more delicate tissue, necessitating new eyes, but that is the least unsettling thing about a ghola.

To date, no ghola has ever been able to remember their past life. This means they look and sound like the person they once were but know none of their old friends or family. So, the Tleilaxu often make them to be used as disposable agents and workers, but only among the Tleilaxu. They can be quite personable and are far from automatons, but they are never who they once were. This makes most people see them as somehow 'soulless'. For this reason, very few people ever request a ghola, but the Tleilaxu have been known to offer them as gifts for a variety of nefarious reasons.

Tleilaxu often make gholas simply because they can; continually looking for a way to bring someone back with their memories intact. Should they manage to unlock the secret of restoring memory to gholas, the ability to bring loved ones back from the dead as they were would make the Tleilaxu unspeakably powerful.

SWORDMASTERS OF GINAZ

The irrelevance of the gun and the laser in the universe necessitated the development of many new approaches to combat. The Sardaukar are deadly fighters, brutal and remorseless and implacable. The Fremen are rumored to produce warriors of great skill, if rarely seen. But few surpass the skill of those trained on Ginaz.

Ginaz is a waterlogged planet in a small, out of the way system ruled over by a relatively small and out-of-the-way House. Its swordmasters, however, are the most talented and deadly wielders of the blade seen in any galaxy.

The school is generally believed to have been founded by the renowned swordsman Jool-Noret. But this is not entirely true. He was a master of 93 fighting techniques and promised to teach his skills when the time was right. Unfortunately, he was killed in his sleep by a freak tidal wave before he ever found the right moment. So, while he never taught a single student, the masters of Ginaz have based their teaching on the object lesson of his story. There is no time like

the present, and skills you never teach will die with you. That is not to say they will teach just anyone, but to the right student, nothing is hidden by these legendary fighters.

GINAZ TRAINING

Capable of taking on multiple foes at a time, single-handedly and remaining confident of victory, each swordmaster of Ginaz undergoes the most rigorous and exhausting of training regimens, designed to inculcate both physical discipline and martial skill. Those sent for training follow a strict regime as they move from island to island on Ginaz. Each island has a different lesson to teach, and those who fail to learn any of these lessons are sent home as failures.

The lessons encompass survival, observation, and even poetry, often before a recruit even picks up a blade. When they move on to actual combat training it is often done in heavy armor or on recalcitrant steeds to ensure the student learns to fight in the most difficult of circumstances.

While the Sardaukar are shaped by their training to follow the Emperor, to serve him above all else, the swordmasters value the art of combat. They practice each aspect of the process of fighting, from the care of the weapon to the delivery of the killing blow. Nothing is left to chance, nothing is unconsidered. It's an intricate process, evolved over generations. Just like the blades the swordmasters use, their every motion is honed to the sharpest of edges.

For this reason, Ginaz-trained swordmasters are in demand throughout the universe, sought after as bodyguards and assassins, or as weapons masters for Major Houses (often with the side duty of training the progeny of the House's leader). If Imperial conditioning is the mark of the ideal doctor, then the training of the Ginaz is the equivalent for anyone privileged enough to wear a sword in the presence of a duke or a baron. Ginaz training does not only concentrate on the use of the weapon, as has been noted, but also on tactics and the strategies of effective fighting and warfare. A skilled Ginaz swordmaster is as useful plotting the grand sweep of a battle as most Mentats, though, obviously, without the same capacity to process information. They are also often dispatched as advanced scouts, to survey the locations their employer is to visit, searching for threats or potential allies.

ART OF WAR

The origins of the Ginaz style of fighting can be traced back to the Butlerian Jihad. The need for reliable and effective martial arts led many to reconsider older, more obscure, traditions of warfare. Old Terran fencing was a significant influence on the formulation of the Ginaz fighting style, utilizing its careful footwork and speed as a means of confusing opponents, tricking them off balance and using this opportunity to pierce their shield with the outstretched blade. By combining ancient approaches to the use of the blade with their understanding of how the new shielding technology functioned, the early Ginaz Swordmasters developed the perfect means of puncturing them. And puncturing the skin of their enemies. Over many years of constant evolution, the style reached an apogee of effectiveness and elegance.

To watch a swordmaster of Ginaz in battle is to witness something both beautiful and horrifying. As they move past their enemy, blood seems to blossom on the enemies' body in several places at once, leaving the wounded shrieking and those less fortunate in ragged heaps.

The pedagogical aspect of the swordmasters is an important one. After all, they are masters, and that word carries a weight of responsibility. The techniques and approaches of the Ginaz need to be communicated and spread, though only to those deemed suitable. Obtaining entrance to one of the Ginaz's most esteemed

academies is not easy and, even for those Ginaz pledging their loyalty to one of the various Landsraad factions, teaching an unworthy pupil is rare. Despite the enormous sums paid by the Major Houses to obtain the swordmasters, few masters would try and force a swordmaster to train someone deemed unsuitable. After all, to lose the loyalty of a swordmaster would be foolish and, potentially, deadly.

THE HONOR OF THE BLADE

While some swordmasters do become assassins, this is relatively rare. It is also rare for a swordmaster to turn on their employer. This is not rendered impossible, in the fashion that imperial conditioning guarantees, but it is part of the Ginaz discipline that one holds to a bond. Training is a pact made with your master, to learn all you can and implement it as effectively and consistently as possible. To do otherwise would be to scorn your master's teaching. The same holds true for an employer. They have trusted you in employing you, and to betray that trust in a cowardly fashion is fundamentally unbefitting a swordmaster.

This approach to refinement and to honor also colors the ways in which the swordmasters fight. It is not enough to simply be effective and highly trained. A swordmaster must fight with flair, with finesse. The end goal might be the enemy's defeat, but to do so in an ugly fashion is to have, in some way, failed. Fighting should never be clumsy, or haphazard. It should be as fluid and elegant as possible. Technique and purpose perfectly aligned.

There are no recorded encounters of a Ginaz swordmaster encountering a large number of Sardaukar, and there is some speculation over whether the individual flair and skill of the swordmaster would eventually fail when placed against the relentless advance of the Emperor's soldier-fanatics. Certainly, it is fair to say that, while the Ginaz are peerless in individual combat, or in those knife-duels so beloved of the Harkonnens, they have never been able to assemble an army. This is, of course, in part because of the size and impecunious nature of Ginaz itself.

Had such a formidable organization originated on a planet with a more fortuitous location, who is to say what it might have achieved? Though, as scholars love to point out, would a planet with greater advantages in terms of political power produce such a martial tradition?

All of this, is, of course, pure conjecture, but it is nevertheless interesting to compare the fact that those places said to have produced the most dangerous and effective fighting styles and forces are harsh backwaters. Whatever the truth, the swordmasters of Ginaz remain amongst the most consistently lethal fighters the universe has ever witnessed. Those who fail to respect their prowess invite their own death. But at least it is sure to be a most stylish death.

PLANETS



The Imperium under the Padishah Empire rules over millions of planets in the Known Universe, which is comprised of worlds inhabited and documented by humanity. The Imperium expanded slowly during the reign of the current Emperor.

The handful of planets following are less than a glimpse of the expansive universe humanity has conquered. The timeline of the planets ends a decade before the Atreides family moves to Arrakis.

As one of the most significant planets of the Known Universe and a focus of *Dune: Adventures in the Imperium*, Arrakis is described in significantly greater detail than the other planets, and the following section delves deeply into its unique environment and denizens.

ARRAKIS

Star System: Canopus

Moons: Two

Habitable: Class III

Noble House: Harkonnen

Primary Export: Spice melange (alcohol, coffee, mining, and politics)

Languages: Galach, Chakobsa (the Fremen tongue)

Locations of Interest: Arrakeen, Carthag, Polar Sink, Sentinel Rock, Sietch Jacurutu, Sietch Tabr, and Windsack

Arrakis. Dune. The desert planet. A hell of sand, dust, and heat. Little grows, the skies are clear of anything resembling a cloud. The only time the horizon becomes obscured is when the vast sandstorms begin to fill the sky, sweeping down upon the unsuspecting with sufficient ferocity to strip flesh from bone, to mulch metal. It is a world without pity, a world without the capacity for weakness or mercy. Those unprepared for its harshness are doomed to die, flesh reddened, mummified by the heat—every drop of liquid drained from their flesh.

On Arrakis, one's own skin becomes a threat, an impediment to survival. Every pore is an enemy—a means by which moisture can escape, costing minutes of survival. Everything that exists on Arrakis is in some way inimical to human life. Almost every animal species is hostile. Even plants, the sparse grass and hardier flowers, are just competitors for water. There is only one reason anyone would choose to live on Arrakis: the spice melange.

THE PLANET ITSELF

Arrakis is a large planet orbiting the third star in the Canopus system. Arrakis is itself orbited by two moons—both renowned for their markings. One bears a shape which resembles that of a large, desert kangaroo mouse, a species which, while not native to Arrakis, has become something of a symbol for the inhabitants of the great desert—the Fremen. The other looks as though a human hand had been pressed into the surface by some ancient god.

The planet has zero precipitation, though the atmosphere is at least breathable by humans—without the need for any respiratory equipment. It has been hypothesized that the oxygen percentage present on Arrakis must be because of the enormous sandworms as well as sand plankton in the open deserts. There is also an absence of widespread plantlife capable of producing the quantity of oxygen which tests of the atmosphere have indicated to be present. There is much speculation over whether the existing environment of Arrakis was natural or, had in some way been engineered. The Imperial Planteologists Pardot and Liet Kynes have both speculated on this possibility.

PLANETARY HABITABILITY CLASSES

Every planet is a unique location with a vast number of differences than others but fall within five classes for our purposes here. The classes are based on human sustainability and availability of water.

- ✧ **Class I:** Cannot sustain human life.
- ✧ **Class II:** Humans can survive with extreme environmental precautions.
- ✧ **Class III:** Humans can survive with environmental precautions.
- ✧ **Class IV:** Humans can exist without protection.
- ✧ **Class V:** Easily sustain human life without preconditions.

THE ENVIRONMENT

Renowned for its extreme aridity and the difficulty of survival on its surface, Arrakis is a desert planet. Vast, rippling tides of sand cover almost the entire planet—only at the northern pole is there anything resembling a basis for building. At the most northern extremity of the planet, there is a vast stone cap. This is large enough and thick enough to build upon, safely and reliably, secure against both the endlessly shifting sands and the ferocious attacks of the sandworms, which pose a constant threat to any major structure or vehicle located on or near the desert. As a result, the northern pole is the center of activity on Arrakis, and it is here that the greatest concentration of the planet's small population is found. The Shield Wall surrounds the northern extent of the northern cap, keeping it safe from the potential influx of sandworms, and providing some protection from the excoriating impact of the sandstorms which can strip rock, metal, and skin in moments.

The planet possesses several extensive mountain ranges which strike up through the otherwise endless desert, where the planet's various tectonic plates shifted and aligned millennia ago. Significantly smaller outcrops of rock are dotted throughout the vast desert and proffer the few sanctuaries from sandworm attack available in the great open expanses of sand.

FLORA

As stated previously, Arrakis lacks any form of precipitation. There is no rainfall and nothing to sustain a varied or abundant plant life. The few examples of plant life are all highly adapted to surviving in the most inhospitable of climates, with extensive root systems sunk deep into the earth. These both anchor the plant in place, even as the dunes relentlessly move and stir, and allow them to gain some access to the few reserves of water the planet

possesses. These are sunk deep into the earth and are only found using specialist mining technology.

While many of these plant species are found on other occupied worlds, there are a number native to Arrakis—specifically, the creosote bush. The creosote bush is widely remarked upon for its obnoxious odor, which repels any creature who might think to feed on it, and with shallow roots (in marked difference to many of the other plant species), enabling it to move with the undulations of the desert. The base of the creosote bush is also highly toxic and capable of quickly killing any creature foolish enough to ignore the plant's rancid scent. Despite its unprepossessing nature, the creosote bush has become an emblem of fortitude and luck to the planet's indigenous inhabitants—the Fremen. One of the very few occasions on which a member of the Fremen might remove a part of their stillsuits when outside of their settlements is when encountering a creosote bush. The Fremen have been observed rubbing their hands with the leaves of the creosote bush—the oils the plant's leaves exude occlude the pores in the skin, resulting in less moisture being lost as a result.

FAUNA

Just as the flora of Arrakis has adapted to surviving in the harshest of conditions, so too has its wildlife. Most of the animals living on the dunes are small, hardy, and need almost no water to survive. They have also adapted to survive on a diet of whatever they can find. Dune is home to many resilient insects and arachnids—particularly scorpions, which despite their poisonous stings and aggressive nature have long formed a key part of Fremen childhood games—and the centipede. The origin of the centipede is uncertain, with many asserting that it was originally native to Old Terra. It is certainly not, however, native to Arrakis and was in fact introduced to the planet by Pardot Kynes.

The kangaroo mouse is one of the more common of Arrakis' mammals and forms an important part of Fremen religious belief. With its small body, long tail, and exceedingly large eyes, the kangaroo mouse (or, as the Fremen dub it, 'Muad'Dib') is also renowned for its ability to leap implausible distances, hurling itself fearlessly across the sand. Perfectly adapted to the desert conditions it inhabits, it can survive without drinking water at all. It is for this reason, amongst others, that the creature has become so important to the Fremen, who see its survival as a mirror of their own.

Other, somewhat larger creatures also make their home in the desert. The desert hawk and desert owl are both common species of bird on Arrakis—frequent sightings of both happen in the established cities at the northern pole, and across the exposed sands. The birds often shelter in the apertures of homes and civic buildings in

the cities, before venturing out over the wasteland to find food. There is little to be found for even the most diligent of hunters, but such is the nature of survival on Dune.

Among the stranger forms of life known to inhabit Arrakis is the sandtrout. These large, amorphous blobs of matter are found in great numbers in the deep desert, buried deep in the sand. As with the scorpions, playing games with the sandtrout is a common activity for Fremen children, who often 'fish' for the odd creatures. The sandtrouts' relationship to the immense sandworms is a closely-guarded secret, known only to the Imperial Planetologist and the Fremen themselves, and few even among their number.

SANDWORMS

Of course, as even the most careless and disinterested student of Arrakis' native fauna knows, the most impressive and dangerous of the planet's inhabitants is the sandworm. These gigantic beasts live beneath the sand, burrowing deep below the surface, where they form vast networks of tunnels. Called 'Shai-Hulud', or 'Makers', by the Fremen, the sandworms course through the planet, posing a terrible threat to any who travel carelessly across the surface of the sand. The largest of the species, found in the deep deserts, often stretch to 400 meters in length and even the most junior is usually capable of swallowing spice mining rigs whole in a few delirious moments of violence. The thick, sand-colored hide of the sandworm is nearly impenetrable and is lined with something like scales.

To all but the Fremen, the sandworms are terrifying monsters. They lurch up from the sand, consuming anything that they encounter, their huge sightless heads splitting into a mouth lined with endless teeth. Drawn by any rhythmic noise on the surface of the sand, sandworms can surge upwards and towards such noises with terrifying speed and, against creatures of such size, there are almost no weapons which can drive them off. Only massive amounts of explosives—perhaps even atomics—are enough to seriously injure or kill the largest of the sandworms, though they remain in the most inaccessible desert reaches, far beyond the range of even the most ambitious spice miners. The sandworm is not simply a creature of limitless destructive power, however. It is a beast whose importance to the planet of Arrakis, and from there, the universe itself, cannot be underestimated.

The sandworms are so-called not simply for their place of residence, but also for their diet. They consume sand in vast quantities, funneling it into their stomachs where it is digested, along with any organic matter such as sandplankton. Sandworms are entirely suited to Arrakis, being able to sustain themselves without the need for any other life forms save themselves and being incapable of existing in any location with